

Test Case ID	Target Function / Operation	Test Input Vector	Expected Output	Actual Output (Legacy Code)	Pass / Fail	Technical Remarks
TC-01	<code>add_item()</code>	Add 10, 20, 30 (Normal)	Items added; Count = 3; Capacity = 5.	Items added successfully.	PASS	Normal addition within initial capacity works.
TC-02	<code>add_item()</code>	Add 6 items (Triggers Resize)	Array capacity doubles to 10; all 6 items retained.	Items added, but old array memory leaked in heap.	FAIL	Dynamic allocation leaks the previous heap memory buffer.
TC-03	<code>printAll()</code>	Print array with 5 items	Displays indices 0 to 4 cleanly.	Displays 5 valid items plus 1 garbage memory value.	FAIL	Off-by-one error (<code>i <= currentCount</code>) reads past the boundary.
TC-04	<code>findItem()</code>	Search for existing target 30	Returns index 2.	Returns index 2.	PASS	Target is found despite the redundant inner loop.
TC-05	<code>findItem()</code>	Search for missing target 999	Returns -1.	Returns -1.	PASS	Non-existent element lookup handles the return safely.

TC-06	<code>remove_item_at()</code>	Delete valid index 2	Item at index 2 removed; array shifts left; Count = 4.	Item removed; array shifts left correctly.	PASS	Functional, but execution logic contains redundant loops.
TC-07	<code>remove_item_at()</code>	Delete negative index -5	Rejects input with an error message; no state change.	Prints "Invalid index!"; state unchanged.	PASS	Negative index check is present.
TC-08	<code>remove_item_at()</code>	Delete out-of-bounds index 99	Rejects input with an error message; no state change.	Crash or corrupts array state due to invalid memory access.	FAIL	Lacks upper-bound check (<code>idx >= currentCount</code>).
TC-09	<code>processMatrix()</code>	Transpose a 3 × 3 matrix	Transposed grid is displayed correctly.	Displays transposed matrix correctly.	PASS	Function works but lacks dynamic dimension support.
TC-10	Memory Cleanup	Program exits normally	Zero memory leaks reported by Valgrind/AddressSanitizer.	Memory leak reported because allocated memory is not released.	FAIL	Missing <code>delete[]</code> cleanup call.